



On microstructure and mechanical performance of AlCoCrFeMo_{0.5}Ni_x high-entropy alloys

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ABSTRACT

A high-entropy alloy system, AlCoCrFeMo_{0.5}Ni_x, was designed based on the AlCoCrCuFeNi alloy, by replacing Cu with Mo in order to improve strength and thermal stability. Ni content was monitored to see its effect on microstructure and properties. The microstructure of AlCoCrFeMo_{0.5}Ni_x alloys consists of B2, FCC, and σ phase which are all multi-element solid solutions. As nickel content increases, the constituent phases change from B2 + σ phases at $x = 0$ to B2 + FCC + σ phases at $x = 2.0$, reflecting Ni is an FCC stabilizer. The room temperature hardness of the present alloy system decreases from Hv 904 at $x = 0$ to Hv 404 at $x = 2.0$. AlCoCrFeMo_{0.5}Ni_x alloys with $x = 0, 0.5, 1.0$, and 1.5 exhibit higher hot hardness level than that of Ni-based superalloys, In 718 and In 718 H, up to 1273 K. Furthermore, they possess the less negative value of softening coefficient (B_{II}) at high-temperature regime as compared with those of In 718, In 718 H, and T-800. This superiority can be explained with the slow diffusion rate of vacancies in the multi-element matrix. The present alloy system has great potential in high-temperature applications.

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1. Introduction

For thousands of years, practical alloy systems have almost based on one major metallic element. Around thirty alloy systems have been commonly used [1,2]. The reason for not using multi-principal elements to produce more alloys is the expected formation of many complex intermetallic compounds which would render alloys brittle in processing and difficult in analysis [3,4]. To overcome this misunderstanding, Yeh et al. [5,6] proposed the concept of “high-entropy alloys” (HEAs), in 1995, and revealed that high entropy effect could enhance the formation of solid solution phases which include multi-element solutions extended from terminal solid solution or intermetallic compounds. As a result, the formation of elemental phases and stoichiometric intermetallic compounds is largely inhibited. This effect thus renders the microstructure simpler to be analyzed and manipulated. HEAs are

defined as alloys having at least five principal elements with concentrations between 5 and 35 at.%. To date, this new alloy concept have attracted considerable research [6–38] and various alloy systems, such as Al–Co–Cr–Cu–Fe–Ni [11,12,14,16,27], Al–Co–Cr–Cu–Fe–Ni–B [7], Al–Cr–Fe–Mn–Ni [20,26], Co–Cr–Fe–Ni–Ti [37], Al–Co–Cr–Fe–Ni–Ti [18,38], Al–Co–Cr–Fe–Mo–Ni [29,30,32,34] and W–Nb–Mo–Ta–V [31,36] have been reported. Many of them have exhibited attractive properties such as high strength, good oxidation and corrosion resistance, superior softening resistance, and well abrasive wear resistance, which are important for industrial applications.

The Al–Co–Cr–Fe–Mo–Ni alloy system [29,30,32,34] has been designed based on previous Al–Co–Cr–Cu–Fe–Ni alloy system extensively investigated [11,12,14,16,27]. The reason why Mo replaces Cu is to overcome the soft Cu-rich phase formation in Al–Co–Cr–Cu–Fe–Ni system and promote the strength at high temperatures in consideration of high melting point and elastic modulus of Mo. In these researches, Fe [29], Co [30], Mo [32] and Cr [34] were varied to see their effects on microstructure and properties. It is noticed that the combination of hardness and toughness is in general superior over Al–Co–Cr–Cu–Fe–Ni alloy system, and moreover, hot hardness and softening resistance are superior to In

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718, In 718 H, and T-800. In this study, in consideration of its strongest FCC forming ability and its importance to contribute high-temperature performance for superalloys, Ni content was varied from $x = 0$ (0 at.%Ni) to $x = 2$ (30.8 at.%Ni) to generate a series of AlCoCrFeMo_{0.5}Ni_x alloys and to see its unique effect on microstructure and properties. The molar ratio of Mo was similarly set to 0.5 for sake of lower cost and easy comparison with former alloy series. It is expected that this study could enhance the whole understanding on Al–Co–Cr–Fe–Mo–Ni alloy system from both scientific and industrial points. In addition, this could help further alloy design to achieve even better performance.

For assessing the strength and softening resistance of the present alloys, in situ hot hardness measurement was performed as for tools, molds and superalloys in literature [39,40]. In analysis of hot hardness data, the equation proposed by Westbrook is used to evaluate the softening resistance and revealing the softening mechanism [39–41]:

$$H = Ae^{-BT} \quad (1)$$

where H = hardness (kg mm⁻²); T = test temperature (K); and A , B = constants. The constant A is called intrinsic hardness and B is softening coefficient or thermal coefficient of hardness. Two sets of A and B constants are typically obtained at low- and high-temperature ranges corresponding to two different deformation mechanisms. At low-temperature region the major deformation mechanism of softening is slip, while at high-temperature region the major deformation mechanisms of softening are related with diffusional phenomenon, such as (1) slip along more than one slip system due to increased atomic mobility, (2) dislocation climb, and (3) grain boundary rotation [42,43].

2. Experimental procedure

The experimental HEAs are AlCoCrFeMo_{0.5}Ni_x with different nickel content from $x = 0$ to $x = 2.0$ as listed in Table 1. For convenience, Ni- x is used to represent AlCoCrFeMo_{0.5}Ni_x. They were prepared by the arc melting and casting method in a protective atmosphere of argon. The raw materials were elemental Al, Co, Cr, Fe and Ni (all purity > 99wt.%) and Fe–61 wt.% Mo master alloy. Four times of re-melting and re-solidification were performed for each sample to assure the uniform composition. These cast ingots were cut and polished for observation under a scanning electron microscope (SEM, JEOL JSM-5410). The chemical compositions of these cast alloys were analyzed by SEM energy dispersive spectrometry. Crystal structures were examined with an X-ray diffractometer (Rigaku ME510-FM2) operated at 30 kV and 20 mA with a scanning rate of 4°/min from 20 to 100°. Room temperature macrohardness and microhardness measurements were made using Mitutoyo HV-115 and HM-115 hardness testing machines with loads of 5 kg and 10 g, respectively. The duration time in each indentation is 15 s. The cracks around each indentation were observed and measured in length by using CCD (charge coupled device) camera of the hardness testing machine (Mitutoyo HV-115). Hot hardness measurement was performed with a high-

temperature hardness tester (Akashi AVK-CO) equipped with a heating and vacuum part (Akashi AVK-HF). A load of 5 kg with a dwell time of 15 s was used. Before measuring, the chamber was purged with argon, and then evacuated to 1.9×10^{-4} torr for three times and finally filled with argon to a pressure close to 1 atm. Besides the measurement at room temperature, the sample was tested at various temperatures with 100 K interval from 573 K to 1273 K. Heating rate was 10 K/min for each temperature increase. The specimens for hot hardness measurements were 1.5 mm in thickness and 10 mm in diameter.

3. Results and discussion

3.1. Microstructure and constituent phases

X-ray diffraction patterns of the AlCoCrFeMo_{0.5}Ni_x alloys in the as-cast state are shown in Fig. 1. Three phases: FCC, B2 (ordered BCC), and σ phase (body-centered tetragonal structure) [29,30,32,34] are identified and listed in Table 1 for comparison. Their lattice constants are listed in Table 2. Their relative amounts can be related with their relative peak intensities. Ni-0 alloy almost consists of B2 phase and have a small amount of σ phase. Ni-0.5 alloy has a large amount of σ phase and thus a reduced amount of B2 phase. Ni-1.0 alloy conversely has increased B2 phase and decreased σ phase. Ni-1.5 alloy has a significant newly-formed FCC phase and the reduced amount of σ phase. Ni-2.0 alloy has a larger amount of FCC phase and both BCC and σ phase are reduced. As compared with the content effect of another FCC stabilizer, Co, which is thought to be the second to Ni in terms of FCC stabilization, the phase partitioning among the three phases is very different [30]. When Co content v is zero (AlCo_vCrFeMo_{0.5}Ni alloys), there is a large amount of σ phase besides B2 phase. As Co content increases from $v = 0.5$ to 1.5, B2 phase is increased and σ phase is decreased. Co content of $v = 2.0$ yields a significant FCC phase besides the BCC phase and thus σ phase is largely reduced. In summary, with increasing Co content, the formation tendency is from σ to FCC whereas that with increasing Ni is from B2 to FCC. Furthermore, higher content of Ni is more effective in forming FCC phase than that of Co.

Fig. 2 shows the SEM backscattered electron images of the as-cast AlCoCrFeMo_{0.5}Ni_x alloys. The chemical compositions of different regions and phases were obtained by EDS and listed in

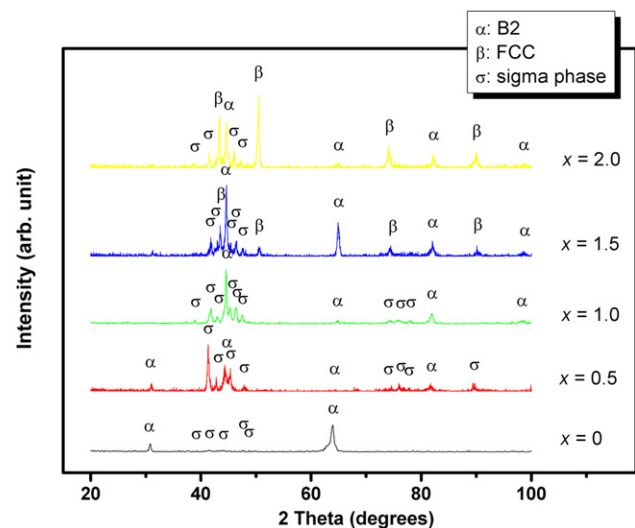


Fig. 1. XRD patterns of AlCoCrFeMo_{0.5}Ni_x alloys with various Ni contents.

Table 1
Compositions of the experimental alloys.

Composition	Designation	Crystal structure
AlCoCrFeMo _{0.5} Ni ₀	Ni-0	B2 + σ
AlCoCrFeMo _{0.5} Ni _{0.5}	Ni-0.5	B2 + σ
AlCoCrFeMo _{0.5} Ni _{1.0}	Ni-1.0	B2 + σ
AlCoCrFeMo _{0.5} Ni _{1.5}	Ni-1.5	B2 + FCC + σ
AlCoCrFeMo _{0.5} Ni _{2.0}	Ni-2.0	B2 + FCC + σ

Table 2
Lattice constants (Å) of different phases in AlCoCrFeMo_{0.5}Ni_x alloys.

Alloy	B2	FCC	σ (BCT) a	σ (BCT) c
Ni-0	2.909	—	8.865	4.640
Ni-0.5	2.882	—	8.849	4.659
Ni-1.0	2.869	—	8.909	4.627
Ni-1.5	2.869	3.602	8.897	4.591
Ni-2.0	2.868	3.614	8.961	4.648

Table 3. Fig. 2 (a) and (b) show the microstructure of Ni-0 and Ni-0.5 alloys, which have polygrain structure without dendritic features. Grain boundaries have thin white phase and grain interiors contain a small amount of white phase. From XRD analysis, it is obvious that matrix phase is B2 and white phase is σ . This is supported by the EDS analysis which shows that white phase is richer in Mo which is one of the two main elements (Cr and Mo) in forming σ phase. Ni-0.5 alloy displays modulated structure like the spinodal decomposition structure as reported for Al_xCoCrCuFeNi alloys [12]. The

characteristic width and spacing in the modulated structure are in the order of submicron meter. The white phase with higher Mo content in the modulated structure is also σ phase whereas the dark phase richer in Al is B2 phase. From the microstructure features, it is believed that Al-rich BCC phase first forms as polygrain structure with some segregation of Mo along grain boundaries during solidification and then σ phase forms in Mo-rich regions and modulated structure forms from BCC phase in the Ni-0.5 alloy during cooling of the solidified ingots.

Fig. 2 (c) shows that Ni-1.0 alloy has a simple dendritic structure composing of dendrite (labeled as A) and interdendrite (labeled as B). From Table 3, it can be seen that the overall composition of dendrites is rich in Al and Ni whereas that of the interdendrites is rich in Cr and Mo. From the morphology, it can be expected that the dendrite is a BCC phase initially, which is a multi-element solid solution derived from the binary NiAl compound in the pseudo-binary phase diagram of (Ni, Co, Fe)–(Al, Cr, Mo), and then precipitates white σ phase to form σ + B2 modulated structure

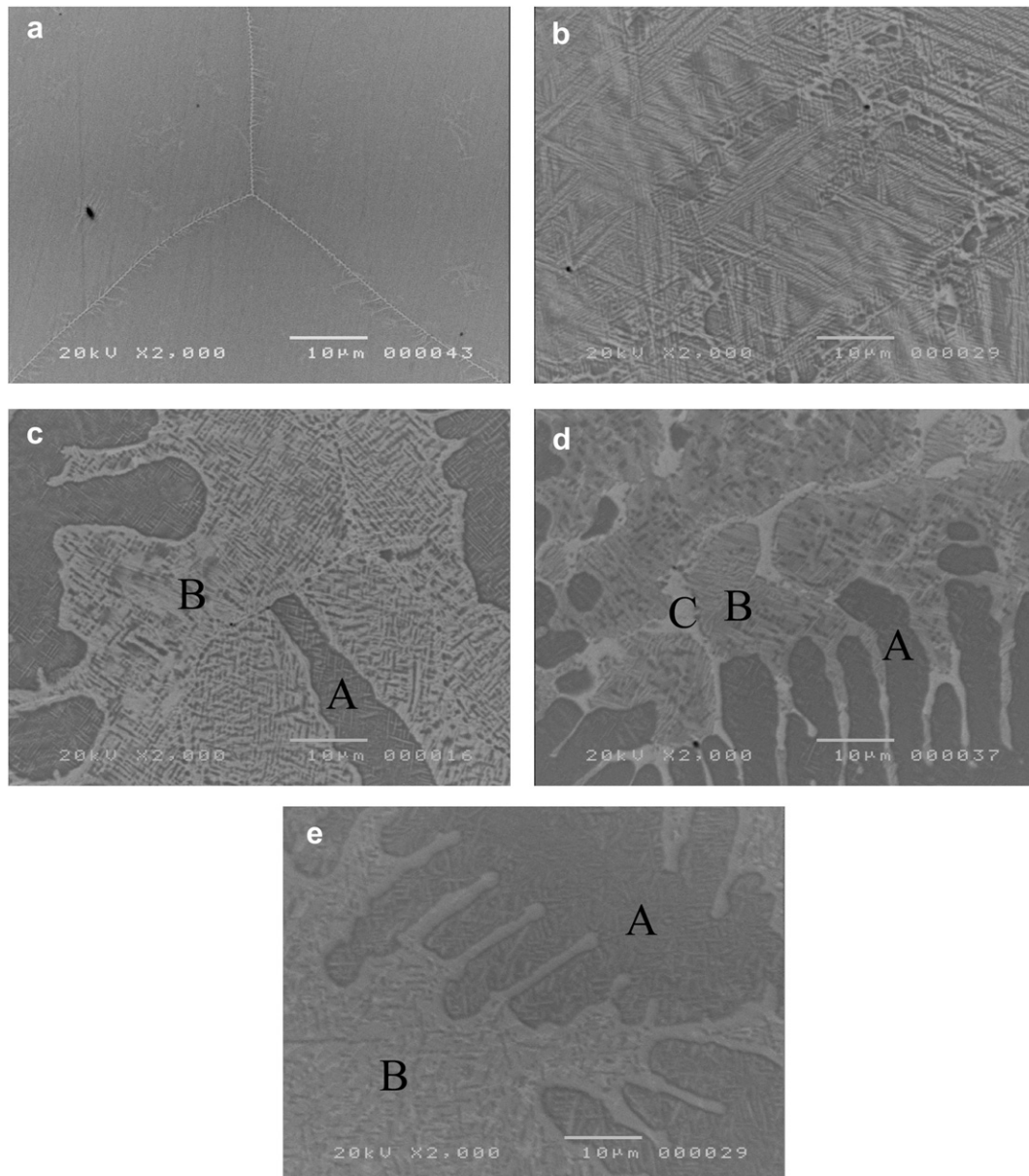


Fig. 2. SEM backscattered electron images of as-cast AlCoCrFeMo_{0.5}Ni_x alloys: x = (a) 0, (b) 0.5, (c) 1.0, (d) 1.5, and (e) 2.0.

Table 3
Chemical compositions of different regions in AlCoCrFeMo_{0.5}Ni_x alloys (at%).

x-value	Region or phase	Al	Cr	Fe	Co	Ni	Mo
0	Nominal	22.2	22.2	22.2	22.2	0	11.2
	Grain	21.8	22.3	21.5	21.6	0	12.8
	Grain boundary ^c	21.7	21.3	19.1	22.7	0	15.2
0.5	Nominal	20.0	20.0	20.0	20.0	10.0	10.0
	Grain	20.7	19.6	19.8	19.7	9.6	10.6
	Grain (white phase) ^c	18.4	19.7	19.5	18.4	9.1	14.9
	Grain (dark phase) ^c	22.7	19.9	19.8	19.0	10.8	7.8
	Grain boundary ^c	14.1	22.4	18.7	18.1	7.7	19.0
1.0	Nominal	18.2	18.2	18.2	18.2	18.2	9.0
	DR ^a (A region overall)	24.5	12.3	15.6	18.1	24.4	5.1
	ID ^b (B region overall)	11.3	21.0	19.3	17.5	14.1	16.8
	Nominal	16.7	16.7	16.7	16.7	25.0	8.2
1.5	DR ^a (A region overall)	27.5	9.2	12.8	15.0	32.7	2.8
	ID ^b (B region overall)	13.2	19.8	19.1	16.7	21.5	9.7
	B region (white phase) ^c	10.9	22.5	18.3	17.1	19.1	12.1
	B region (dark phase) ^c	21.4	14.2	17.8	15.6	26.5	4.5
	C region (white phase) ^c	5.3	25.4	17.3	15.9	12.2	23.9
	Nominal	15.4	15.4	15.4	15.4	30.8	7.6
	DR ^a (A region overall)	25.0	9.4	12.1	12.7	36.6	4.2
	A region (white phase) ^c	21.2	11.9	13.6	13.1	34.1	6.1
2.0	A region (dark phase) ^c	26.0	8.4	10.8	12.7	39.0	3.1
	ID ^b (B region overall)	14.0	16.7	16.7	15.2	28.0	9.4
	B region (white phase) ^c	10.4	17.3	18.0	16.2	29.1	9.0
	B region (dark phase) ^c	17.1	14.2	15.2	14.5	32.1	6.9

^a DR: dendrite.

^b ID: interdendrite.

^c Values are approximate because the phase or region size detected is below the instrument resolution, i.e., <1 m.

during the cooling process after solidification. This is because the formation of σ phase (rich in Cr and Mo) leads to higher Al and Ni contents in the matrix and causes the formation of B2 phase.

Due to the partitioning of more Cr and Mo to liquid phase during solidification, the Cr- and Mo-rich interdendrite, being the outer envelope of the cored dendrite, is also BCC phase initially, but precipitates the white σ phase in the cooling stage after solidification. Because of the difference in Cr and Mo contents, BCC-dendrite precipitates a smaller volume fraction of fine σ phase but BCC-interdendrite precipitates a larger volume fraction of σ phase.

TEM investigation has been conducted on the modulated structure of Ni-1.0 alloy. Fig. 3 shows the bright-field image and selected area diffraction (SAD) patterns of dendrite. Table 4 lists TEM-EDS compositions of gray and white phases. The gray phase (corresponding to the white phase in SEM image based on the EDS analysis) is lesser in volume fraction than the white phase. SAD pattern of [010] zone axis confirms its BCT structure with $a = 8.93$ Å and $c = 4.62$ Å. Its composition can be written as (Cr, Mo)_{44.9}(Fe, Co, Ni, Al)_{55.1}. As for Al- and Ni-rich white phase which is extended from NiAl binary intermetallic compound, SAD pattern of $\bar{1}\bar{1}13$ confirms its B2 structure with $a = 2.89$ Å. Fig. 4 shows the bright-field image and SAD patterns of interdendrite. Gray phase is more than white phase in volume fraction. This is opposite to their relative volume fraction in the dendrite. Similar observation is obtained in Fig. 2(c) except the phase color is reversed. The difference has been attributed to the higher Cr and Mo contents of interdendrite than dendrite. SAD pattern of [010] zone axis in gray phase also confirms its BCT structure with $a = 8.92$ Å and $c = 4.58$ Å. Its composition can be written as (Cr, Mo) 50.6(Fe, Co, Ni, Al) 49.4. As for white phase, SAD pattern of $\bar{1}\bar{1}13$ confirms its B2 structure with $a = 2.88$ Å.

According to phase identification in Fig. 1, phase morphology in the microstructure of Ni-1.5 and Ni-2.0 alloys in Fig. 2 and the composition partition in Table 3, it is expected that (Al, Ni)-rich dendrite (labeled as A) is initially a BCC phase at high temperatures and then transforms into σ phase plus B2 phase during

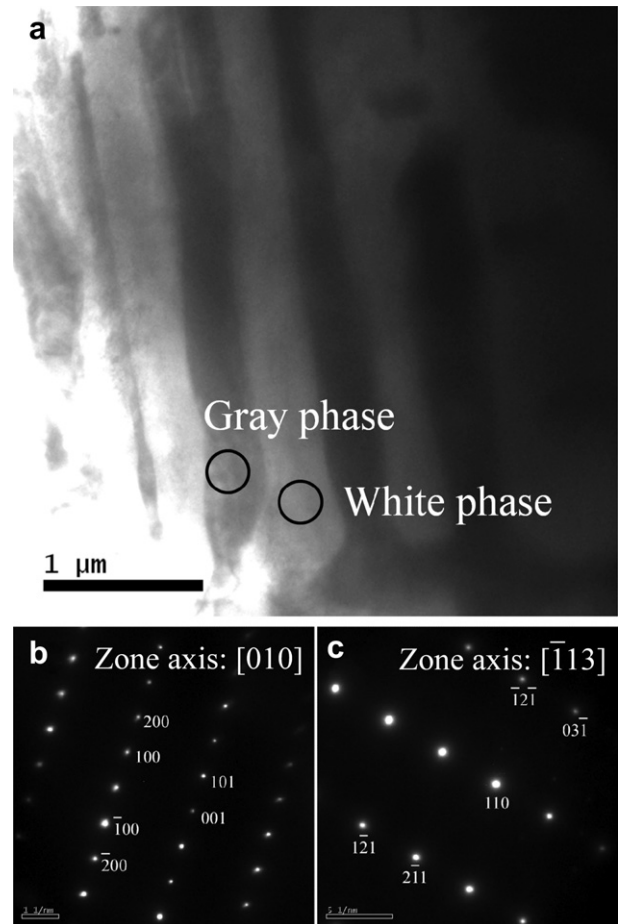


Fig. 3. TEM microstructure of the dendrite region in the Ni-1.0 alloy: (a) the bright field image, (b) SAD pattern of gray phase, (c) SAD pattern of white phase.

cooling. However, the interdendrite (labeled as B) initially assumes FCC structure (gray) and then partially transforms into σ phase (white) plus B2 phase (dark) due to its high Ni and low Al contents. In the Ni-1.5 alloy it is further found that a final solidified region between the interdendrite has even higher segregation of Cr and Mo and also forms as σ phase after solidification.

3.2. Mechanical properties

3.2.1. Room-temperature hardness and crack length of indentation

Fig. 5 plots the room-temperature Vickers hardness and overall crack length around the indent marks of as-cast AlCoCrFeMo_{0.5}Ni_x alloys with various. Those of the Al_xCoCrCuFeNi alloys [11] from which the present alloy were derived are compared to see the property improvement. The variation of hardness as shown in Fig. 5 (a) exhibits a strong correlation with the relative volume fractions of hard σ phase, hard B2 phase, and soft FCC phase and the compositions of these constituent phases. It can be seen that Ni-0 alloy displays the highest hardness of Hv 904. Because this

Table 4
TEM-EDS compositions of different phases in the Ni-1.0 alloy.

Phase	Al	Co	Cr	Fe	Mo	Ni
Gray phase in dendrite region	7.6	17.7	28.3	20.3	16.6	9.5
White phase in dendrite region	28.0	19.5	6.6	13.9	0.6	31.4
Gray phase in interdendrite region	3.0	18.3	32.6	23.8	17.0	5.3
White phase in interdendrite region	33.9	16.7	5.0	13.3	0.8	30.3

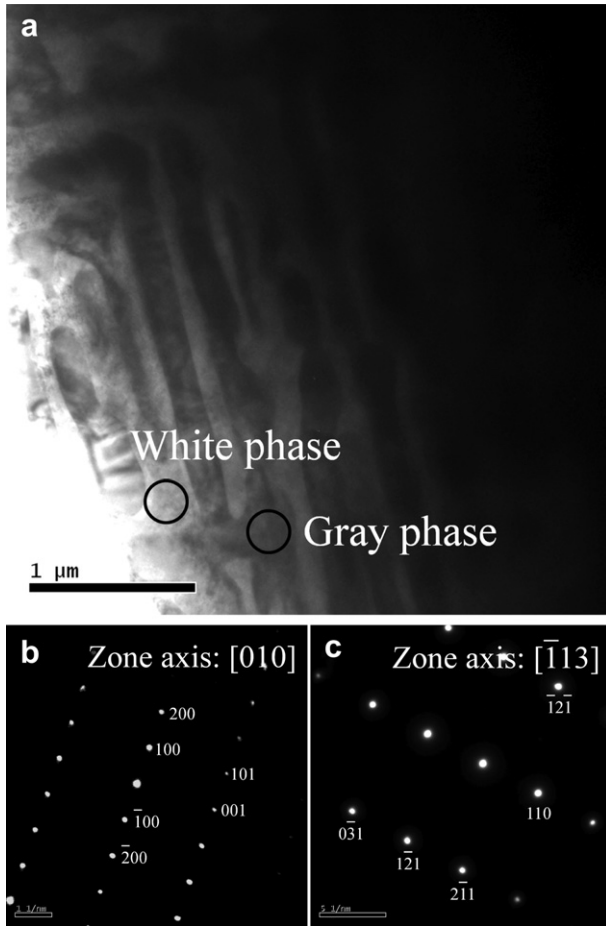


Fig. 4. TEM microstructure of the interdendrite region in the Ni-1.0 alloy: (a) the bright field image, (b) SAD pattern of gray phase, (c) SAD pattern of white phase.

alloy is B2 phase except a small fraction of σ phase, this demonstrates that B2 phase with the composition $\text{AlCoCrFeMo}_{0.5}$ (without Ni) is very hard. Ni-0.5 and Ni-1.0 alloys are similar in hardness, both around Hv 770. But, as Ni-0.5 alloy has a larger amount of σ phase than Ni-1.0 alloy and microhardness measurements on the dendrite (with less σ phase) and the interdendrite (with more σ phase) in Ni-1.0 alloy give Hv 634 and Hv 824, respectively, B2 phase in Ni-0.5 alloy is harder than σ phase whereas σ phase in Ni-1.0 alloy is harder than B2 phase. The relative hardness change of these two phases implies that Ni content in both phases has a different influence on hardness (see EDS analyses of dark B2 and white σ phases of Ni-0.5 alloy in Table 3). More

addition of Ni lets B2 phase become softer and lets σ phase become harder. Compared with Ni-1.0 alloy, Ni-1.5 alloy has a lower hardness of Hv 627 and Ni-2.0 alloy has an even lower hardness of Hv 404. This softening is attributable to the fact that the soft FCC phase increases in amount and B2 phase becomes softer due to more Ni and lesser Mo involved.

For the variation of crack length which is an indicator of fracture toughness [44], it can be found from Fig. 5 (a) that except the crack lengths of Ni-0 alloy and Ni-1.0 are 66 and 13 μm , respectively, Ni-0.5, Ni-1.5 and Ni-2.0 are all zero. Comparing with the crack length and hardness of the $\text{Al}_x\text{CoCrCuFeNi}$ alloys [11], the present alloys are superior in hardness and fracture toughness. For example, the hardest Ni-0 alloy not only has a crack length 66 μm much less than the hardest $\text{Al}_x\text{CoCrCuFeNi}$ alloys from $x = 2.3$ to 3.0, but also has a hardness of Hv 904 much higher than the latter alloys ranging from Hv 603 to Hv 644. Furthermore, Ni-0.5 alloy having a hardness of Hv 770 has no crack around indent marks whereas the best alloy $\text{Al}_{2.0}\text{CoCrCuFeNi}$ has a combination of crack length, 20 μm , and hardness, Hv 566. This clearly suggests that the present alloy system which has different combination of B2, σ and FCC phases displays much better toughness at much higher hardness as compared with $\text{Al}_x\text{CoCrCuFeNi}$ alloys. This indicates that modulated structure of $\text{BCC} + \sigma$ is superior over that of $\text{BCC} + \text{B2}$. From the toughening mechanism of composites, interfaces between different crystal structures could deflect the crack and retard its propagation. On the other hand, $\text{BCC} + \text{B2}$ modulated structure could be regarded as a crystal since both have almost the same lattice constant. Coherent interface has no deflecting effect on crack propagation. It should be mentioned that although σ phase is in general detrimental to the ductility of stainless steels and superalloys, a large volume fraction of fine σ interconnected with fine B2 phase as seen in the present alloys could provide an excellent combination of hardness and toughness. This is important for those structural applications which require high hardness and proper toughness [30].

3.2.2. Hot hardness

Fig. 6 shows the variations of hot hardness (H) with temperature (T) for the $\text{AlCoCrFeMo}_{0.5}\text{Ni}_x$ alloys, Ni-based superalloys of In 718 (as-received) and In 718 H (precipitation hardened), and Co-based superalloy of Tribaloy T-800 (data are obtained from Stellite Company website). It can be seen that T-800 has much higher hardness in lower temperature range but rapidly softening at above 800 K. Precipitation hardened In 718 H have medium hardness. Nonhardened In 718 has the lowest hardness but becomes hardened at above 900 K due to precipitation hardening. It is interesting to see all these three commercial alloys approach to the same hardness at 1100 K and soften at the same rate. In contrast, except Ni-2.0, all alloys have significantly higher hot hardness and lower

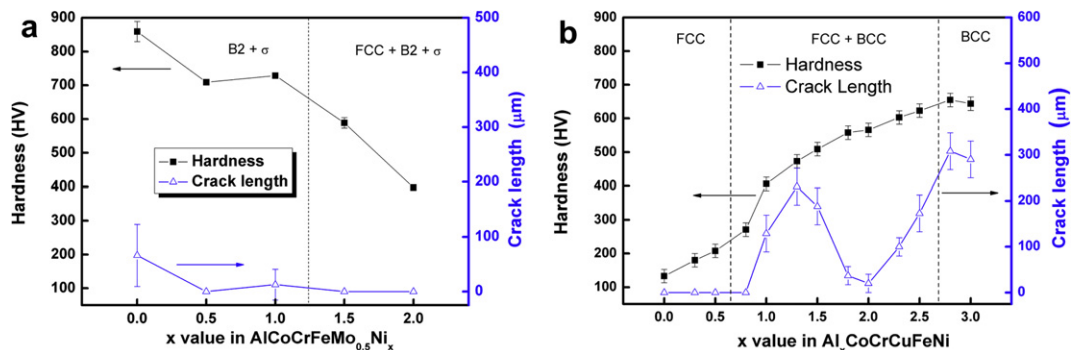


Fig. 5. (a) Vickers hardness and total crack length of $\text{AlCoCrFeMo}_{0.5}\text{Ni}_x$ alloys as functions of Ni content; and (b) those of $\text{Al}_x\text{CoCrCuFeNi}$ alloys for comparison with (a) [11].

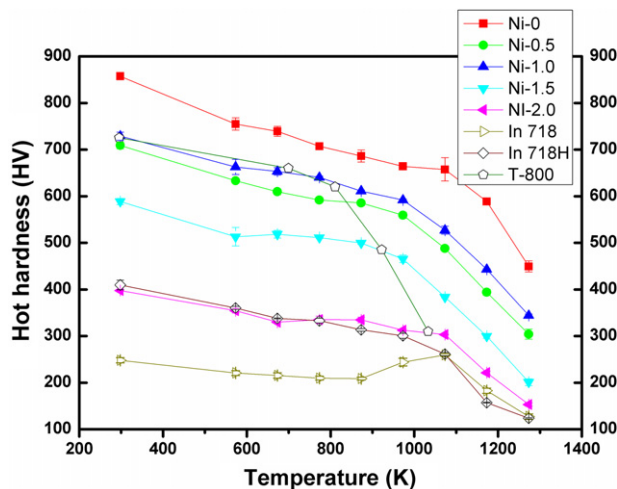


Fig. 6. Hot hardness vs. temperature plots for $\text{AlCoCrFeMo}_{0.5}\text{Ni}_x$ alloys ($x = 0, 0.5, 1.0, 1.5$, and 2.0), In 718, In 718 H and T-800 alloys from room temperature to 1273 K.

softening rate. Although Ni-1.5 and Ni-1.0 alloys have lower hardness than T-800 in lower temperature range, lower softening rate lets them maintain much higher hardness than T-800 at above 1000 K. It is noted that Ni-0 alloy possesses the highest hot hardness of about Hv 450 (i.e. around Rc 45) at 1273 K, which is around 3.5 times hardness values, $\sim \text{Hv } 127$, of In 718 and In 718 H.

Fig. 7 is a re-plot of Fig. 6 by using $\ln(H)$ vs. T . For each alloy, two straight lines based on Westbrook equation (see Eq. [1]) are used to fit the experimental curve. At their intersection temperatures is suggested a change of deformation mechanism [41] as mentioned in the Introduction section. The values of transition temperature (T_T), intrinsic hardness (A), and softening coefficient (B) calculated according to Westbrook [41] are summarized in Table 5. It can be seen that Ni-0 alloy has the highest transition temperature of 1100 K, which is much greater than that of T-800 by around 300 K and slightly higher than that of In 718 and In 718 H. In addition, the present alloys in general possess high softening resistance as reflected by the lower softening coefficient B_{II} .

The reasons why the present alloys have higher transition temperature and higher softening coefficient B_{II} need further discussion. For this, the softening coefficients of pure metals such

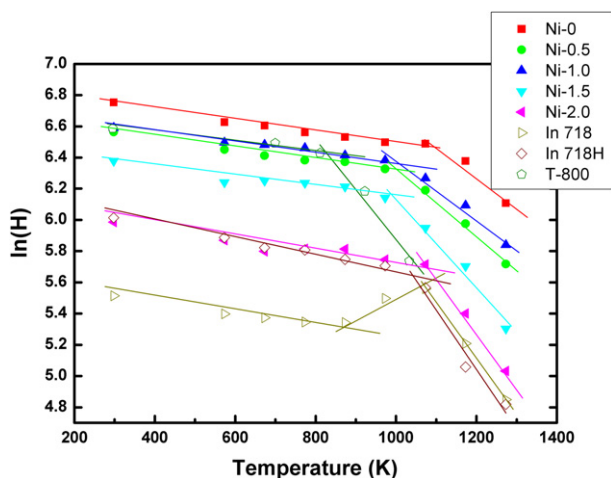


Fig. 7. $\ln$ (hot hardness) plotted against temperature (from room temperature to 1273 K) for $\text{AlCoCrFeMo}_{0.5}\text{Ni}_x$, In 718, In 718 H, and T-800 alloys (data are obtained from the Stellite Company website).

Table 5

Transition temperature (T_T), intrinsic hardness (A) and softening coefficient (B) of $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys and compared superalloys [41].

Composition	Transition temperature (T_T) K	Below T_T		Above T_T	
		A_I	B_I	A_{II}	B_{II}
Ni-0	1100	935	$-3.47\text{E-}4$	5201	$-1.90\text{E-}3$
Ni-0.5	994	776	$-3.40\text{E-}4$	4214	$-2.04\text{E-}3$
Ni-1.0	995	796	$-2.99\text{E-}4$	3528	$-1.82\text{E-}3$
Ni-1.5	986	634	$-3.02\text{E-}4$	7190	$-2.77\text{E-}3$
Ni-2.0	1078	430	$-3.26\text{E-}4$	11940	$-3.41\text{E-}3$
T-800 ^a	818	791	$-2.84\text{E-}4$	8041	$-3.12\text{E-}3$
In 718 ^b	1069	268	$-3.10\text{E-}4$	11766	$-3.55\text{E-}3$
In 718 H ^c	1037	487	$-5.29\text{E-}4$	13778	$-3.74\text{E-}3$

^a Composition of T-800: Co: 47.6%; Mo: 28%; Cr: 18%; Si: 3.4%; Fe: 1.5% and Ni: 1.5%.

^b Composition of In 718: Ni: 53%; Cr: 19%; Fe: 18.5%; Nb: 5.1%; Mo: 3%; Ti: 0.9% and Al: 0.5%.

^c Annealed at 1253 K for 30 min, air cooled, and aged at 1033 K for 8 h, furnace cooled at a rate of 55 K, heated to 893 K for 8 h, air cooled.

as Mo, Ni, Cr, Si, and Fe [45] are compared with the present data. It can be found that they have two-order magnitude larger than those of the present $\text{AlCoCrFeMo}_{0.5}\text{Ni}_x$ alloys and the superalloys. This demonstrates that the alloying indeed has tremendous effect for lowering the softening coefficient. For the region below transition temperature where the major deformation mechanism of softening is slip, the softening resistance is basically related to the decreasing rate of Young's modulus with increasing temperature and thus displays similar B_I value. However, in stage II, the deformation is enhanced by vacancy diffusion through dislocation climb and grain boundary sliding mechanisms, and thus might differ with each other if vacancy diffusion rate is different. As $\text{AlCoCrFeMo}_{0.5}\text{Ni}_x$ alloy system consists of B2, FCC, and σ phase, which are all multi-element solid-solution phases, it is expected vacancy diffusion rate is lower than that in superalloys since vacancies in each above phase are surrounded by more solute atoms of different elements and thus easily trapped by them in order to lower lattice strain energy around vacancies. As a result the migration and diffusion rate of vacancies are lower and provide better resistance to creep deformation as compared with the case with a host element. From the above comparison, the present alloys from Ni-0 to Ni-1.5 thus have great potential in high-temperature applications which require higher strength and softening resistance.

4. Conclusions

As Ni content increases, the crystal structure of $\text{AlCoCrFeMo}_{0.5}\text{Ni}_x$ HEAs changes from B2 + minor σ phase at $x = 0$ to B2 + FCC + σ phase at $x = 2.0$. All these three phases are multi-element solid solutions. Ni-0 alloy composed of B2 phase has the highest hardness of Hv 904. As Ni content increases, alloy hardness decreases: Hv 770 at $x = 1.0$, Hv 627 at $x = 1.5$, and Hv 404 at $x = 2.0$. $\text{AlCoCrFeMo}_{0.5}\text{Ni}_x$ alloys are superior in hardness and fracture toughness over the $\text{Al}_x\text{CoCrCuFeNi}$ alloy system from which it derives. This demonstrates that modulated structure of BCC + σ is superior over that of BCC + B2. All $\text{AlCoCrFeMo}_{0.5}\text{Ni}_x$ alloys except Ni-2.0 have significantly higher hot hardness and lower softening rate. All alloys have higher hardness than the commercial Ni-based superalloys In 718 and In 718 H at 1273 K. In particular, Ni-0 alloy exhibits the highest hot hardness of about Hv 450, which is around 3.5 times hardness values, $\sim \text{Hv } 127$, of In 718 and In 718 H. Ni-0, Ni-0.5, Ni-1.0, and Ni-1.5 alloys have the least negative value of softening coefficient above transition temperature (B_{II}) as compared to those of In 718, In 718 H and T-800, suggesting that these alloys could sustain its strength up to 1273 K very effectively. The excellent softening resistance is explained with the lower migration and

diffusion rate of vacancies in the present alloys as compared with that in superalloys. The $\text{AlCoCrFeMo}_{0.5}\text{Ni}_x$ alloy system has great potential in those high-temperature structural applications which require high hardness, softening resistance, and proper toughness.

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